

LAB 2 Ring Oscillator

Technology: 22nm PTM-HP Nominal VDD = 0.8 V

k is selected as 1.3 based on Lab1 (K is ratio of PMOS Width to NMOS Width within the inverter subcircuit)

HSPICE simulation for 21 inverter ring oscillator, 3 inverter ring oscillator and 1 inverter ring

Lab 2.0

$N = 2 \times \text{number of inverters}$

Metric	21 inverters	3 inverters
T (period)	131.53ps	17.71ps
f	7.6GHz	56.45GHz
tpHL	3.13ps	2.851ps
N*tpHL	42*tpHL = 131.63ps	6*tpHL = 17.11ps
tpLH	3.18ps	3.04ps
N*tpLH	42*tpLH = 133.60ps	6*tpLH = 18.24ps

Lab 2.25

Frequency vs. VSS

gnd_vss was swept from 0 V to 0.4 V in 10mV steps (VDD held fixed at nominal 0.8 V), reducing the supply headroom seen by the inverters, for both the 21-inverter and 3-inverter rings. Source data: `results_21/results.mt0` (ring.sp) and `results_3/results.mt0` (ring_3.sp).

Frequency drops monotonically as VSS is raised — from 7.60GHz to 0.33GHz (21-inv, VSS = 0-0.39V) and from 56.45GHz to 2.14GHz (3-inv, VSS = 0-0.4V).

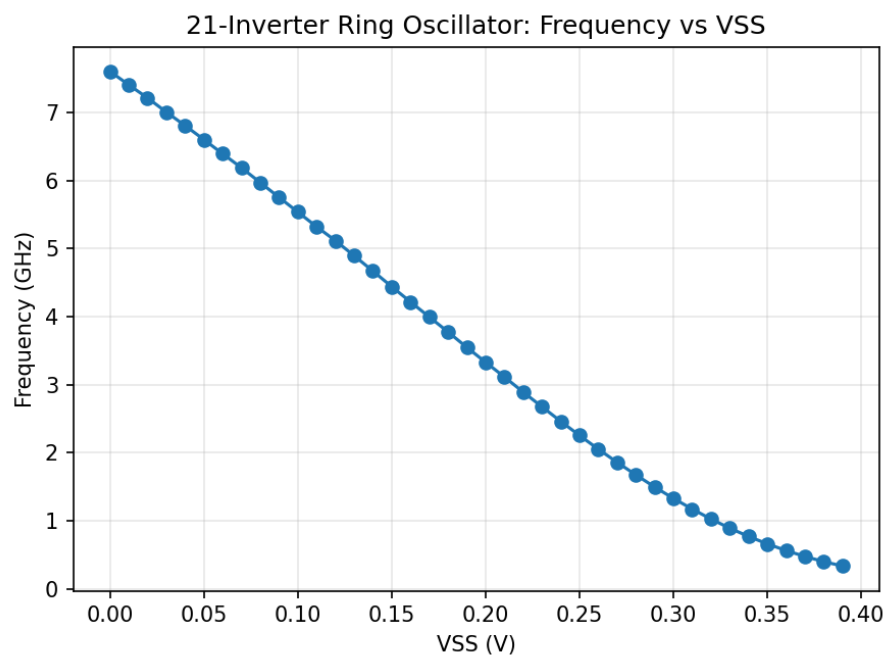


Figure 1: 21-inverter ring: frequency vs VSS

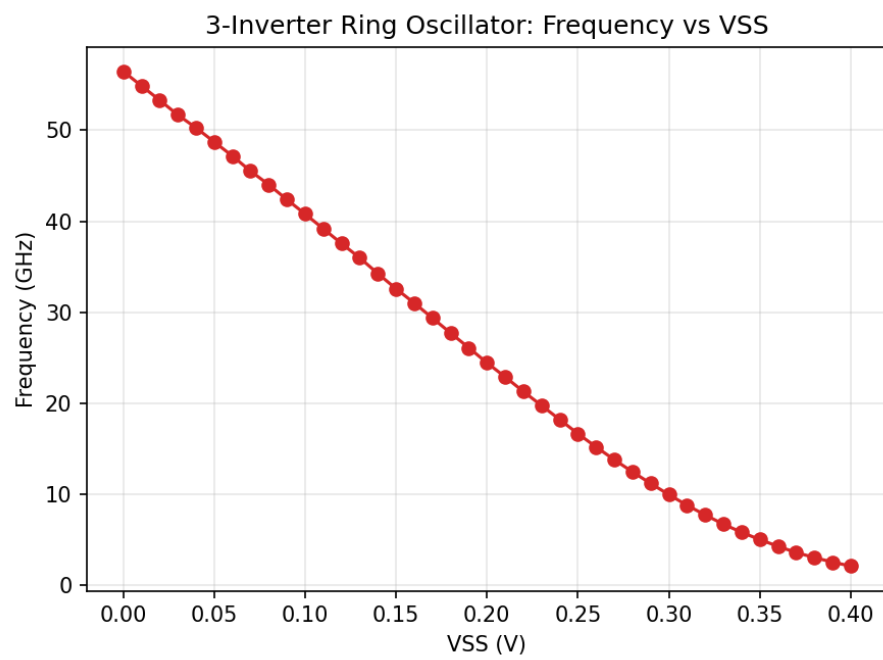


Figure 2: 3-inverter ring: frequency vs VSS